1. Fibre: highly transparent core of index ,
with circular core radius ; cladding index . A
semiconductor laser , on the axis extension, emits a
half-angle beam. To admit it all, the fibre end has
a sintered cap of radius , of the core material.
As shown, cladding is omitted. is a point source.
The fibre is in air of index , taken as .

For all rays to reach the cap, what condition must
distance from source to satisfy?

For , - distance , let the axis
angle be . For at , , , can
each ray travel in the fibre by total internal reflection?